

Question number	Answer	Additional guidance	Mark
6(a)	methionine leucine isoleucine tyrosine (1)	ACCEPT met leu iso tyr combinations of names and abbreviations	(1)

Question number	Answer	Additional guidance	Mark
6(b)(i)	Base number 3 becomes cytosine (C) substitution Base number 6 becomes number 5 in the sequence deletion	All three correct = 2 marks Any one or two correct = 1 mark IGNORE point mutation throughout IGNORE swapping mutation IGNORE elimination / frameshift	(2)

	Base number 9 becomes number 10 in the sequence insertion	IGNORE addition / frameshift	
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Question number	Answer
*6(b)(ii)	Indicative content: Substitution: • only affects one triplet codon • may not change the amino acid • e.g. number 6 becomes A, would still code for leucine • may change amino acid • e.g. number 1 becomes G, resulting in valine • may result in a stop codon • e.g. number 12 becomes G Deletion: • one base removed will shift the reading frame back one place • all amino acids after the mutation will be affected • closer to the start of the gene the greater the affect • fewer amino acids coded for • e.g. remove base 4 and sequence becomes methionine serine phenylalanine threonine Insertion: • one base added will shift the reading frame forward one place • all amino acids after the mutation will be affected • closer to the start of the gene the greater the affect • e.g. add C between numbers 9 and 10 and sequence becomes leucine proline serine

Aspects to comment on:

1. Substitution changing the amino acid
2. Deletion changing sequence
3. Insertion changing sequence
4. Stop codons appearing shortening the sequence
5. Substitution may have no effect
6. Position of {insertion / addition} significant

Level 1

1 mark : correct statement about mutations

2 marks : 1 aspect commented on with a corresponding illustration **OR** 2 or more aspects commented on but no illustrations

Level 2

3 marks : 2 aspects commented on with corresponding illustrations **OR** 3 or more aspects commented but only 1 or 2 illustrations

4 marks : 3 aspects commented on with corresponding illustrations

Level 3

	5 marks : 4 aspects commented on with corresponding illustrations 6 marks : 5 aspects commented on with corresponding illustrations
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